X – Linked Problems

1. Red-green colour blindness is a sex-linked recessive trait. A woman with normal vision whose father was colour blind marries a man with normal vision. What is the genotype of each of these people? What can you predict about their children?

2. Haemophilia is a disease caused by a recessive trait on the X chromosome. You are a genetic counsellor. Bob and his sister Mary come to your office. Bob is 25 and has no children. Mary is 23 and engaged. Bob and Mary are both normal as are both their parents. However, they had a brother and an uncle who died of haemophilia. Bob and Mary both want to know what their chances are of having haemophiliac children. What would you say to them?

3. A certain form of muscular dystrophy is inherited as a sex-linked, recessive gene. Jack has muscular dystrophy. (Neither of his parents has this disease.) Jane, Jack's wife, does not have muscular dystrophy, but her father does. What fraction of their daughters would you expect to have muscular dystrophy? What fraction of their sons would you expect to have muscular dystrophy?

4. Colour-blindness is a sex-linked recessive gene. A colour-blind woman marries a man with normal vision. She is pregnant. What is the chance that her child will be (a) a girl with normal vision? (b) a colour-blind girl? (c) a boy with normal vision? (d) a colour-blind boy?

5. (a) Why are men never heterozygous for an X-linked trait? (b) Why must men always inherit an X-linked trait from their mothers? (c) Can a colour-blind father pass this allele on to his sons? Explain. (d) Can a normal male ever have a daughter that is colour-blind? Explain.

6. Anthony, a colour-blind person got married, divorced, and remarried. He and his first wife, Linda, have a colour-blind son, a colour-blind daughter, a normal son and a normal daughter. He and his current wife, Mary, have
four boys and two girls, none of whom are colour-blind. Give the genotype of each wife and state whether or not she is colour-blind.